POWER

U1 LM7812CT
U2 LM7912CT_C7432211
U5 LM7805CT_C725349
J8, J7, J6
C1, C2, C3, C4, C15, C27, C33, C34, C35, C16, C17
+12AC, −12AC, GND, +12, −12V, +5V

Controller

U12 ESP32-S3-DEVKITC-1-N8R2
U17, U18, U19
GPIO 8, GPIO 16, GPIO 38, GPIO 17, GPIO 18, GPIO 12, GPIO 9, GPIO 1, GPIO 2
DGND

Function Genarator

U20 MCP4131-104E/P
U8 MCP4131-104E/P
U9 TL071CP
Q2 TIP31C, Q3 TIP32C
J9 BNC7T-J-P-XX-RA-BH1
J5 Connector_Socket
J12 AD9833_ALT
R29, R31, R3, R13, C42, C43, C7, C8
Amplitude, GPIO 15, GPIO 13, GPIO 11, GPIO 10, GPIO 14

PWM

R2, R1, Q4, Q1 2N3904
J10, J11
C6, C5
GPIO 21, GPIO 47
MF1/4W-1KΩ±1% T52
BNC7T-J-P-XX-RA-BH1

Oscilloscope

U6 OPA4192IPWR
C20, C22, C_Small, −12V
CH1_BUF, CH2_BUF, CH3_BUF, CH4_BUF
IND+, INC+, INA+, INB+
R7, R9, R8, R21, R6, R5, R11, R23, R12, R24, R10, R22
MF1/4W-100KΩ±1% T52
BNC, J4, J3, J2, J1
C19, C18, C37, C40
BAT54S_C545549
D6, D5, D7, D8

Ref

REF3025AIDBZT
U10
C36 C_Small
TLV9062IDR
U11A, U11B
C21 C_Small
2.5V, ref

ESP32-S3 ADC INPUT PROTECTION (CH1 – CH4)

R32, R33, R34, R35 1k
D9, D10, D11, D12
BAT54S

Designed by mars0x1

4-Channel 10MHz DSO | 12.5MHz DDS Function Gen | Dual Pulse Gen
Powered by Hack Club

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